

Section: CHS48.3x3.2 Span: 4.00 m

PASS

Design Inputs

Span L = 4.00 m
Section: CHS48.3x3.2 (CHS)
Grade: Grade 350 fy = 350 MPa
k_x = 1.00 Le_x = 4.00 m
k_y = 1.00 Le_y = 4.00 m
N* = 0.0 kN (factored, all combos)
Self-weight UDL (G) = 0.035 kN/m
Live-load category: Office (psi_c = 0.4, psi_l = 0.4)
Deflection limit: L/360

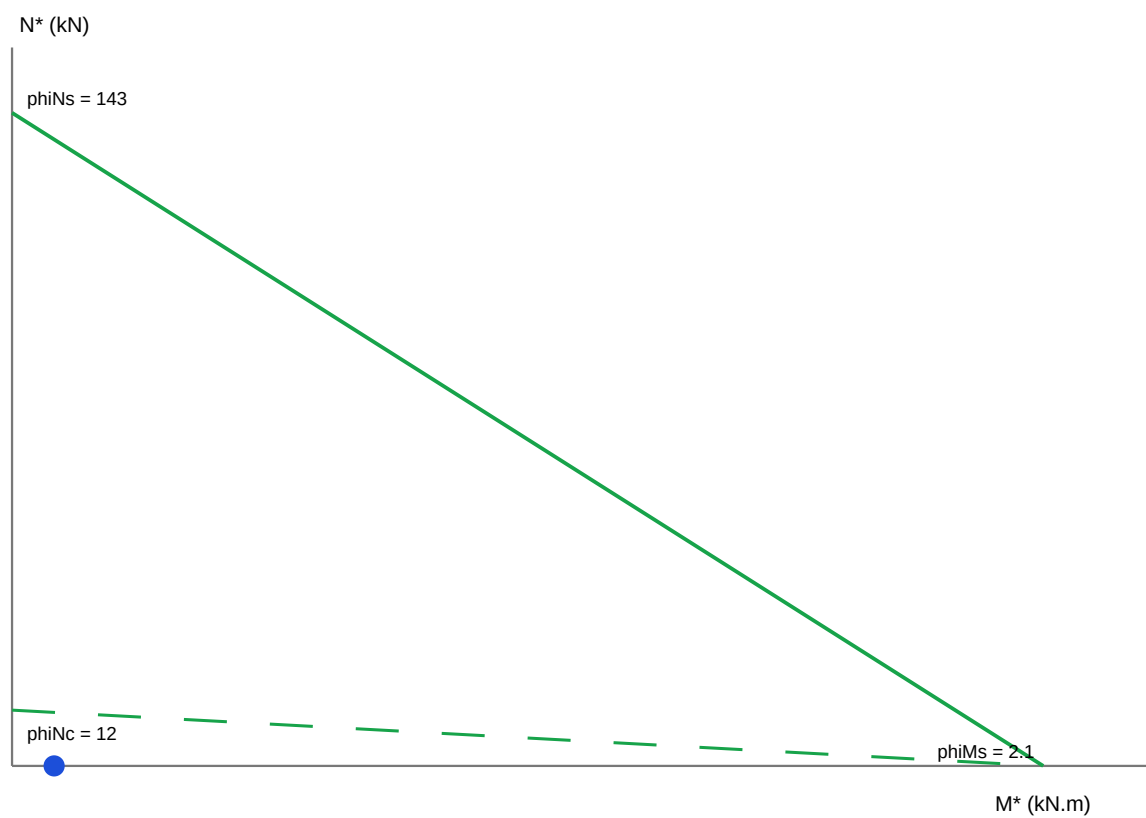
Section Properties

Designation CHS48.3x3.2
Mass 3.6 kg/m
Ag 453.00 mm^2
Ix 1.17e+5 mm^4
Iy 1.17e+5 mm^4
rx 16.1 mm
ry 16.1 mm
Ze_x 6520.00 mm^3

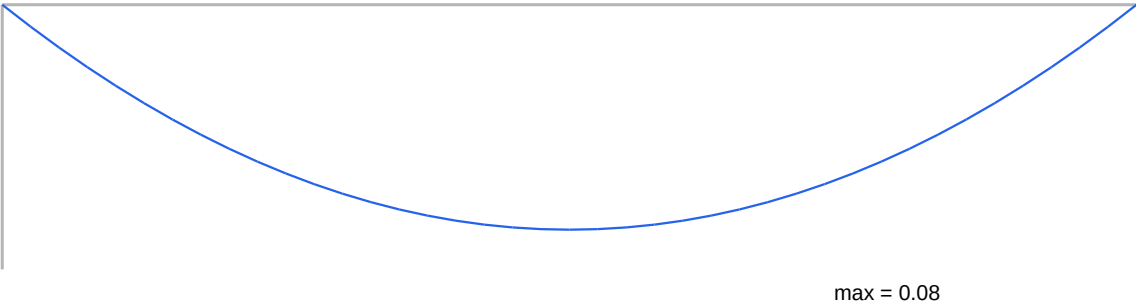
Capacity Check Summary

Check	Demand	Capacity	Util	Status
Axial - Section (phiNs)	0.0 kN	142.7 kN	0.0%	PASS
Axial - Member x (phiNc_x)	0.0 kN	12.2 kN	0.0%	PASS
Axial - Member y (phiNc_y)	0.0 kN	12.2 kN	0.0%	PASS
Bending x - Section (phiMsx)	0.08 kN.m	2.05 kN.m	4.1%	PASS
Combined - Section (Cl. 8.3.3)	0.041	<= 1.00	4.1%	PASS
Combined - Member in-plane (Cl. 8.4.2)	0.041	<= 1.00	4.1%	PASS
Combined - Member out-of-plane (Cl. 8.4.4)	0.041	<= 1.00	4.1%	PASS
Self-weight deflection	4.97 mm	11.11 mm	44.8%	PASS

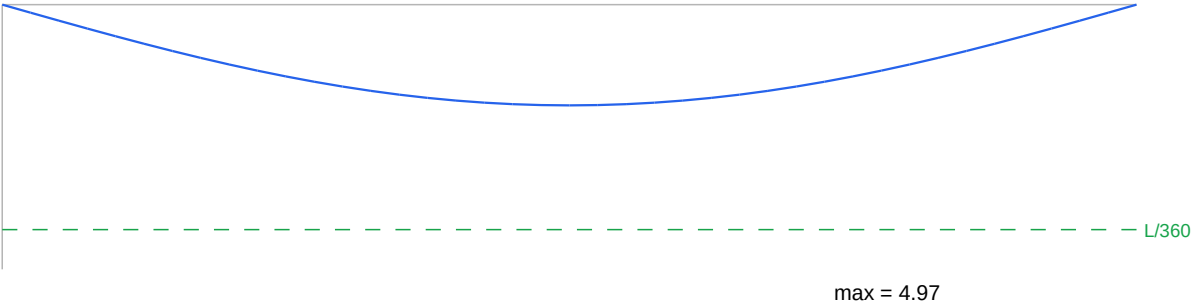
N*-M*x Interaction Diagram (AS4100 Cl. 8.3/8.4)



Bending Moment Diagram - 1.2G + Wu + Ec·Q (kN.m)



Self-weight Deflection Profile (mm)



1. Section classification

[AS4100 Cl. 5.2.2]

$f_y = 350 \text{ MPa}$ (AS1163 flat - closed section)
 Section class: compact $Z_{e_x} = 6520.00 \text{ mm}^3$

2. Axial member capacity

[AS4100 Cl. 6.3.3]

$\phi N_s = 0.9 \times k_f \times A_g \times f_y / 1000 = 142.7 \text{ kN}$ [0.0%]
 $\lambda_{n_x} = (L_{e_x} / r_x) \times \sqrt{k_f \times f_y / 250} = 294.5$; $\alpha_{c_x} = 0.085 \rightarrow \phi N_{c_x} = 12.2 \text{ kN}$
 $\lambda_{n_y} = (L_{e_y} / r_y) \times \sqrt{k_f \times f_y / 250} = 294.5$; $\alpha_{c_y} = 0.085 \rightarrow \phi N_{c_y} = 12.2 \text{ kN}$
 $\phi N_c = \min(\phi N_{c_x}, \phi N_{c_y}) = 12.2 \text{ kN}$

3. Bending capacity

[AS4100 Cl. 5]

$\phi M_{sx} = 0.9 \times Z_{e_x} \times f_y / 1e6 = 2.05 \text{ kN.m}$
 $\phi M_{bx} = \phi M_{sx} = 2.05 \text{ kN.m}$ - no LTB (closed section)

4. Bending actions

[AS1170.0 Cl. 4.2.2]

Self-weight UDL (G) = 0.035 kN/m
 Combos: 1.2G+1.5Q | 1.2G+Wu+ ψ_c ·Q ($\psi_c = 0.4$) | 0.9G+Wu
 Governing: $1.2G + Wu + \dot{E}_c \cdot Q \rightarrow M^* = 0.08 \text{ kN.m}$ [4.1%]

5. Combined actions

[AS4100 Cl. 8.3 / 8.4]

Section (Cl. 8.3.3): $N^* / \phi N_s + M^* / \phi M_{sx} = 0.041 \leq 1.0 \rightarrow \text{PASS}$
 Member in-plane (Cl. 8.4.2): $N^* / \phi N_c + M^* / \phi M_{sx} = 0.041 \leq 1.0 \rightarrow \text{PASS}$
 Member out-of-plane (Cl. 8.4.4): $N^* / \phi N_c + M^* / \phi M_{bx} = 0.041 \leq 1.0 \rightarrow \text{PASS}$

6. Self-weight deflection

[AS4100 Cl. 1.5.3]

$\delta = 4.97 \text{ mm} \leq L / 360 = 11.11 \text{ mm} \rightarrow \text{PASS}$

7. Overall result

[AS4100]

Overall: PASS